PIPELINING MECHANICS Instruction-Level Parallelism in the CPU Based on Andrew S. Tanenbaum, Structured Computer Organization – Section 2.1.5, pp. 65–67

Core Definition Pipelining is a form of instruction-level parallelism that splits 5 instruction execution into several sequential hardware stages, so multiple instructions are in different stages of completion at the same moment. stages of hardware working in parallel on the classic pipeline (Fig. 2-4) Instruction-level parallelism (within one instruction stream) is one of two general forms of parallelism Tanenbaum identifies – the other being processor-level parallelism (multiple CPUs). 2

Architectural Blueprint The five-stage pipeline (Tanenbaum, Fig. 2-4a) S1 Instruction S2 Instruction S3 Operand S4 Instruction S5 Write

Fetch	Decode	Fetch	Execution	Back
Fetches instruction from memory into a Determines instruction type and Locates operands from registers or Runs operands through the datapath Writes the result to the destination				
	operands			